UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which input pattern stops when the user enters -1?
   A. A fixed for loop
   B. A function prototype
   C. A switch statement
   D. A sentinel-controlled loop

2. Which statement immediately exits a loop?
   A. break
   B. return 0
   C. cin
   D. continue

3. Which item is NOT a programming paradigm?
   A. Object-oriented programming
   B. Structured programming
   C. Debugging
   D. Procedural programming

4. Which declaration creates a floating-point variable named price?
   A. bool price;
   B. int price;
   C. float price;
   D. char price;

5. Which part of a for loop is executed after each iteration?
   A. The loop body twice
   B. The update expression
   C. The initialisation
   D. The condition only

6. Which phase of the program development life cycle defines what the program must do?
   A. Coding
   B. Maintenance
   C. Analysis
   D. Testing

7. Which loop is guaranteed to execute its body at least once?
   A. for
   B. if
   C. do...while
   D. while

8. What does the % operator return?
   A. The absolute value
   B. The quotient
   C. The square root
   D. The remainder of integer division

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9. What is the index of the first element of a C++ array?
   A. The array size
   B. 1
   C. -1
   D. 0

10. Which values can a bool variable hold?
   A. true or false
   B. any integer
   C. any character
   D. any decimal

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(3 marks)

b) Explain THREE types of programming errors and give one example of each.

(2 marks)

QUESTION 2

A concert charges RM15 for ages 0-12, RM25 for ages 13-17 and RM40 for ages 18 and above. A negative age is invalid.

a) Draw a flowchart for the ticket price selection.

(6 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(7 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

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PART C (30 MARKS)

QUESTION 1

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER